

# BSU Cyber Squad Camp 2026

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**July 13 – 17, 2026**

Monday – Friday | No prior experience needed



## ARRIVAL / CHECK-IN

8:45 – 9:25 a.m.

## CAMP HOURS

9:25 a.m. – 3:35 p.m.

## PICK-UP

3:35 – 4:15 p.m.

## LOCATION

Room 113B, CS Building

*Buffet lunch provided daily*

## WHAT YOU'LL LEARN

1

**Mon, Jul 13**

Python basics and how the internet works

2

**Tue, Jul 14**

CIA triad, passwords, and account security

3

**Wed, Jul 15**

Cryptography: ciphers, hashing, and TLS

4

**Thu, Jul 16**

Web security: SQL injection, XSS, OWASP

5

**Fri, Jul 17**

CTF competitions, blue team, and careers

## DAILY SCHEDULE

### Time

### Session

### Slides

9:25 – 12:05

**Lectures & Python exercises**

Parts 1, 2, 3, 6

12:05 – 12:55

*Lunch*

12:55 – 1:45

**Cyber Range labs**

Part 4

1:45 – 2:35

**Interactive game exercises**

Parts 5, 7

2:35 – 3:25

**Pair work & presentations**

Part 9

3:25 – 3:35

**Wrap-up**

Part 8

## WHAT YOU NEED

### Laptop + Browser

Everything runs in the cloud - no software installation needed

### Google Colab

Free Python notebooks - sign in with any Google account

### US Cyber Range

Lab credentials provided on Day 1 for all five environments

**Day 1**

Python &amp; Internet

**Day 2**

Passwords

**Day 3**

Cryptography

**Day 4**

Web &amp; AI

**Day 5**

CTF &amp; Blue Team

**SCHEDULE**

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**Wrap-up**

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**SESSION DETAILS****Lectures (9:25 – 12:05)**

Get oriented and learn Python basics. Part 1: Orientation. Part 2: Linux, networking, CLI, SSH. Python exercise intro and Cyber Range walkthrough (Parts 3, 6).

**Cyber Range (12:55 – 1:45)**

Cyber Basics (2025.7) — VM login, CLI navigation, SSH into remote host.

**Interactive Game (1:45 – 2:35)**

Part 5 — Network packet simulation game. Students build and inspect packets.

**Pair Work (2:35 – 3:25)**

Day 1 coding challenge in Day1\_student.ipynb. Work with your partner on CLI scripts.

**Wrap Up (3:25 – 3:35)**

Recap Linux and networking. Preview Day 2: passwords and hashing.

**KEY CONCEPTS****Python**

Variables, loops, functions

**CLI**

Linux command-line navigation

**SSH**

Secure remote shell access

**OSI Model**

7-layer network stack

**TCP/IP**

Internet protocol suite

**Subnetting**

Dividing IP address space

**Port**

Numbered network endpoint

**Firewall**

Network traffic filter

**Day 1**

Python &amp; Internet

**Day 2**

Passwords

**Day 3**

Cryptography

**Day 4**

Web &amp; AI

**Day 5**

CTF &amp; Blue Team

**SCHEDULE**

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Part 9

3:25 – 3:35

**Wrap-up**

Part 8

**SESSION DETAILS****Lectures (9:25 – 12:05)**

Security foundations: CIA triad, threat actors, authentication (Part 2).  
Password entropy math, keyspace calculation, Python exercise (Parts 3, 6).

**Cyber Range (12:55 – 1:45)**

Cyber Basics (2025.7) — Password cracking, hash comparison, salt demo.

**Interactive Game (1:45 – 2:35)**

Part 5 — Password strength game and threat scenario quiz.

**Pair Work (2:35 – 3:25)**

Build a keyspace calculator and hash cracker in Day2\_student.ipynb.

**Wrap Up (3:25 – 3:35)**

Recap CIA triad and hashing. Preview Day 3: cryptography.

**KEY CONCEPTS****CIA Triad**

Confidentiality,  
Integrity, Availability

**Hashing**

One-way fingerprint  
function

**Salt**

Random bytes added  
before hashing

**Entropy**

Measure of password  
randomness

**Auth Factor**

Something you  
know/have/are

**Threat Actor**

Who carries out an  
attack

**Risk**

Likelihood x Impact

**AES**

Advanced Encryption  
Standard

**Day 1**

Python &amp; Internet

**Day 2**

Passwords

**Day 3**

Cryptography

**Day 4**

Web &amp; AI

**Day 5**

CTF &amp; Blue Team

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**SESSION DETAILS****Lectures (9:25 – 12:05)**

History of cryptography and symmetric ciphers. Caesar, Base64, XOR (Part 2). RSA walkthrough, TLS handshake, frequency analysis, Python exercise (Parts 3, 6).

**Cyber Range (12:55 – 1:45)**

Kali Linux (2025.7) — Capture and decrypt traffic, frequency analysis challenge.

**Interactive Game (1:45 – 2:35)**

Part 5 — Cipher challenge game and frequency analysis puzzle.

**Pair Work (2:35 – 3:25)**

Implement Caesar cipher, XOR encoder, and Base64 decoder in Day3\_student.ipynb.

**Wrap Up (3:25 – 3:35)**

Recap cipher types and TLS. Preview Day 4: web security.

**KEY CONCEPTS****Caesar**

Shift-based substitution cipher

**Base64**

Binary-to-text encoding

**XOR**

Bitwise exclusive-or cipher

**RSA**

Public-key asymmetric crypto

**TLS**

Transport Layer Security

**Freq. Analysis**

Breaking ciphers by letter count

**Public Key**

Freely shared encryption key

**Certificate**

Signed proof of identity

## Day 1

Python &amp; Internet

## Day 2

Passwords

## Day 3

Cryptography

## Day 4

Web &amp; AI

## Day 5

CTF &amp; Blue Team

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## SESSION DETAILS

**Lectures (9:25 – 12:05)**

Web app architecture, URLs, client-server model. SQL injection and XSS (Part 2). OWASP Top 10, session hijacking, AI prompt injection, Python exercise (Parts 3, 6).

**Cyber Range (12:55 – 1:45)**

Kali Linux (2025.7) — SQLi and XSS on a vulnerable web app, Burp Suite intro.

**Interactive Game (1:45 – 2:35)**

Part 5 — OWASP attack and defend simulation game.

**Pair Work (2:35 – 3:25)**

URL parser, input validator, and prompt injection guard in Day4\_student.ipynb.

**Wrap Up (3:25 – 3:35)**

Recap OWASP Top 10 and AI attacks. Preview Day 5: CTF.

## KEY CONCEPTS

**SQL Injection**

Malicious SQL in user input

**XSS**

Script injected into web page

**OWASP Top 10**

Most critical web vulnerabilities

**Session Cookie**

Token proving you are logged in

**Burp Suite**

Web security testing proxy

**Prompt Inject.**

Hijacking LLM instructions

**WAF**

Web Application Firewall

**CSRF**

Cross-Site Request Forgery

**Day 1**  
Python & Internet**Day 2**  
Passwords**Day 3**  
Cryptography**Day 4**  
Web & AI**Day 5**  
CTF & Blue Team

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## SESSION DETAILS

**Lectures (9:25 – 12:05)**

CTF categories, OSINT methodology, blue team roles. Incidents, logs, OSINT (Part 2). Log analysis, cert roadmap, Python exercise, career paths (Parts 3, 6).

**Cyber Range (12:55 – 1:45)**

Kali Linux (2025.7) — Mini CTF: crypto, web, and forensics challenges.

**Interactive Game (1:45 – 2:35)**

Part 5 — Analyze login logs, identify the breach, write an incident report.

**Presentations (2:35 – 3:25)**

5 pairs, approximately 10 minutes each, or 10 pairs, approximately 5 minutes each: present your week project to the group.

**Wrap Up (3:25 – 3:35)**

Certificates, celebration, and cybersecurity career resources.

## KEY CONCEPTS

**CTF**

Capture the Flag  
competition

**OSINT**

Open Source  
Intelligence

**Blue Team**

Defenders of a  
network

**IOC**

Indicator of  
Compromise

**SIEM**

Security event log  
aggregator

**Forensics**

Recovering digital  
evidence

**Incident Resp.**

Structured breach  
response

**CompTIA Sec+**

Entry cybersecurity  
certificate